

First Use Guide

Build, simulate and compare MBSE / MBPLE models step by step

What this workbench does

It turns a mission into a connected early engineering model: stakeholder needs and objectives lead to requirements; functions and components describe the solution; parameters, verification and simulations provide evidence; configurable alternatives can be compared and approved. It supports preliminary programme and architecture work, not certified detailed-design analysis.

Two synchronized ways to work

Architect view One guided question at a time, with answer guidance, examples, progress and readiness.

Modeler view Dashboard plus direct tables, graphs, matrices, formulas, variability, analysis and delivery.

Both views update the same canonical project; switching never creates a second model.

Choose a scope

Architecture definition	Traceable product and industrial-system architecture
Architecture and simulation	Architecture plus KPI evidence for the current model
Trade-off	Configurable 100% alternatives, comparable runs and a decision

For domain experts who are new to the website, MBSE and MBPLE.

MBSE / MBPLE Workbench - application v1.8.0 - schema 14

Guide refreshed 25 September 2026. The introduction occupies this single page.

Continue with the existing guide, then use the appended answer handbook for every Architect question group.

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Click any entry to jump to that part, chapter or detailed subsection. The same hierarchy is available in the PDF viewer's bookmarks panel.

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Architect view - guided model creation

Architect view builds the same model as Modeler view, but it asks one plain-language question at a time. Your confirmed answers create the model in the background, and you can switch views without losing work.

Choose a perspective

Perspective	Use it for	Effect
Architect view	Guided creation and recap through one question at a time.	Confirmed answers write canonical project content.
Modeler view	Direct editing of tables, matrices, graphs, formulas and analysis workspaces.	Edits use the same project and can mark guided answers for review.

Work through the questions

1	Select the analysis scope Choose Architecture definition, Architecture and simulation, or Trade-off study. The choice controls which later questions appear.
2	Answer one question Type or select the requested information, then click Continue. The answer appears in the progress panel and creates or links the corresponding model content.
3	Use Skip only for a visible gap Skip has no timer and never advances automatically. A skipped required answer remains visible and prevents the study from being reported as Ready.
4	Return and revise safely Use Back or the progress panel to reopen an answer. If a change affects generated content, review the impact message before applying it.

5

Resume later

The current question and confirmed answers are stored with the project. Save Project for a portable JSON backup; loading the project resumes the guided session.

Question order shared by all scopes

Stage	What you provide	What is created
Study setup	Scope, project name and mission.	Project context and the main purpose of the study.
Stakeholder intent	All stakeholders, then the system of interest, needs and objectives.	Mission and stakeholder traceability.
Use cases and requirements	Use cases; quantitative requirements; parameter, operator, target and unit; engineering values where needed.	Need/objective -> requirement traces, formulas, bindings and satisfaction links.
Product architecture	Functions, components, realizations and relevant satisfaction evidence.	Use-case behavior and product technical architecture.
Industrial architecture	Process functions, equipment, product flows, durations and resources.	Industrial behavior, sequence, components and resource evidence.
Review	Confirm traceability, requirements and architecture recaps.	A concise completeness record and visible unresolved gaps.

Questions added by scope

Scope	Additional guided stages	Final recap
Architecture definition	None beyond the shared architecture path.	Overview, stakeholder traceability, architecture and requirements.
Architecture and simulation	KPI selection/formulas, missing engineering inputs and simulation.	Shared recaps plus current simulation evidence.
Trade-off study	Decision question immediately after requirements; common/alternative characteristics; variability, configurations, simulations, comparison and decision.	Shared and simulation recaps plus comparison and baseline.

Requirement satisfaction rule

Every requirement must be satisfied by at least one function, technical component or parameter. Individual functions and components do not each need to satisfy a requirement.

Recap and synchronization

The final pages summarize the study purpose, stakeholder-to-evidence chain, architecture, requirement state and - when relevant - simulation and trade-off evidence. Detailed formulas, matrices and element records remain available in Modeler view.

Review after direct editing

If the model is changed in Modeler view, affected guided answers are marked for review instead of silently remaining current.

How to use this companion

Keep this guide beside the website. Start with the example project once, then create a new project and repeat only the chapter for your chosen scope. Each chapter is complete on its own.

What the website is for

It turns knowledge about a product and the system that produces it into a connected model. The connections let the website check requirements, collect parameters, calculate KPIs, simulate alternatives and retain the evidence behind a decision.

Before you start

- Write down the decision or question you want the model to answer.
- Collect the main actors, needs, objectives, use cases, requirements, product parts, manufacturing steps, equipment and resources.
- Collect each known number with its unit, source and uncertainty. Mark estimates honestly.
- Choose one scope. Do not build all three workflows inside one project unless you intentionally need them.
- Use the sample project as a worked example. Use New project when you are ready to start from scratch.

Website map

Workspace	What you do there	Why it matters
Model	Create elements, relations, requirements and architecture.	This is the connected source of truth.
Variability	Define features, rules, mappings, configurations and 100% realizations.	This turns one 150% family model into valid variants.
Parameters and KPIs	Enter engineering values and define KPI formulas.	This gives simulations reproducible inputs and measures.
Simulation	Select a model and KPIs, validate inputs and run.	This records results, warnings and their source values.
Architecture Trade Study	Compare feasible alternatives and approve a decision.	This separates requirement feasibility from weighted preference.
Export	Save complete JSON or export selected data and reports.	This makes the work portable and reviewable.

MBSE / MBPLE
Program workbench

ACTIVE PROJECT
Configurable Coffee Machine Product Line

Configurable Coffee Machine Product L | + New project | Duplicate | [Trash]

Save Project | Load Project | 150% Product-Line Model

Local autosave: Saved | Revision 1 | Reset sample | Reset all

Snapshots and Delivery Stage C

Create basic local recovery points and export validated, scoped engineering evidence.

Preliminary engineering estimate — not a verified detailed-design result.

Save / Load Project | Snapshots | Selective Export | PDF Report | Demo Checklist

Save Project

Download the complete active project as a portable, validated JSON package.

Configurable Coffee Machine Product Line
Schema 9 · revision 1 · 41 model elements

The project file includes model, variability, configurations, derivations, simulation evidence, Trade Studies and decisions. Application-level snapshots and interface preferences remain local and are not embedded.

Download project file

Load Project

Validate a saved project file before loading it. Invalid input changes nothing and remains available for raw download.

SAVED PROJECT FILE

Choose File | No file chosen

LOAD BEHAVIOR

☒ **Load as new project**
Keep the active project and resolve only a top-level project-ID collision.

☐ **Replace active project**
Preserve the active project ID and create a safety snapshot first.

Validate and load

The project menu provides Save Project, Load Project, revision and local autosave status.

The model chain in plain language

Question	Element or relation to create
Why does this work exist?	Mission -> stakeholder -> need / objective
What must happen?	Use case -> product function and process function
What must be true?	Need or objective -> requirement
What realizes the behavior?	Product function -> product component; process function -> industrial system component
What resources are needed?	Industrial system component -> requiresResource -> resource
How is it measured?	Parameters and KPI definitions linked to the architecture

Relation direction matters

Create relations in the direction shown. In particular, allocate resources to the industrial system components that realize the process functions. Do not attach resources directly to functions.

Chapter 1 - Architecture building

Use this scope when the result is a connected product architecture and industrial system architecture. No simulation or alternative comparison is required.

Start the project

1	Create a clean project Open the project selector, choose New project, give the project a clear name and set the scope to Architecture building. Save a project file immediately as a starting checkpoint. Done when: The dashboard shows the Architecture building path.
2	Use the dashboard as the route map Open each dashboard step in order. Keep the status panel visible. A later step can expose missing earlier connections, so return to fix them instead of bypassing the warning. Done when: The recommended next activity matches the step you are working on.

The screenshot shows the MBSE / MBPLE Programme workbench interface. On the left is a dark sidebar with navigation options: Dashboard, PROBLEM SPACE (2/2), SOLUTION SPACE (2/2), WORKSPACES (Model, Variability, Parameters and KPIs, Simulation, Architecture Trade Study, Export). The main area displays the 'Configurable Coffee Machine Product Line' project. At the top, there's a header with project name, a dropdown menu, and buttons for '+ New project', 'Duplicate', and a trash icon. Below this are 'Save Project' and 'Load Project' buttons, and a dropdown showing '150% Product-Line Model'. A status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'. The 'PROJECT WORKFLOW' section shows the project title and a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' Below this, the 'Overall scope' section allows selecting a scope: 'Architecture building' (selected), 'Architecture building + simulation', or 'Trade study'. A 'Status' bar shows 'Not started', 'In progress', 'Complete', and 'Blocked'. The 'Problem Space' section (2/2 complete) lists two steps: '1 Define mission, system and intent' and '2 Define use-case and requirement scope'.

Architecture building has two problem-space steps and two solution-space steps.

Problem space - Step 1: define mission, system and intent

1.1	Create the mission In Model > Mission and Context, create a Mission. Use one sentence that states the outcome, operating context and beneficiary.
1.2	Create stakeholders Create the people, organizations or external systems that care about the outcome. Flag exactly one stakeholder as the system of interest. Link the mission to the stakeholders with hasStakeholder.

1.3 Create needs and objectives

For each stakeholder, create Needs for expected outcomes and Objectives for measurable targets. Link them from the stakeholder.

Why: Requirements must later be derived from this intent, so the model can explain why they exist.

Done when: At least one mission is linked to stakeholders; exactly one is the system of interest; at least one need and objective exist.

Problem space - Step 2: define use-case and requirement scope

2.1 Create use cases

Describe observable interactions or services, not internal components. Link the involved stakeholders with involvedIn. Ensure every selected use case involves the system of interest.

2.2 Select the working scope

Choose the use cases that belong to this investigation. Keep unrelated future behavior outside the scope.

2.3 Create system requirements

Write one testable statement per requirement. Add a formula where the rule is numeric. Link each requirement from the Need or Objective using derives.

Done when: The selected use cases involve the system of interest and at least one scoped need or objective derives a requirement.

Good requirement

State the subject, required condition, measurable threshold and unit. Example: The machine total mass shall be less than or equal to 1000 kg.

A requirement formula binds a stable model parameter to a comparison such as `@equipment_mass <= 1000`.

Solution space - Step 1: build product functional and technical architecture

3.1	Create product functions For each selected use case, create the product behavior needed to deliver it. Link the use case to each function with hasFunction. Split only when a function has a distinct responsibility or parameter.
3.2	Create product components Create the physical or logical product elements that realize the functions. Use parent-child structure for assemblies and parts when useful.
3.3	Allocate behavior Link every product function to one or more product components with realizedBy. Add satisfies links from requirements to the functions or components that provide the evidence. Done when: Every scoped use case has product functions and every product function is realized by a product component.

Solution space - Step 2: build industrial system functional and technical architecture

4.1	Create process functions Model the manufacturing, inspection, transport and support behavior. Link each relevant use case to process functions with hasFunction.
4.2	Create industrial system components Create the equipment, workstation, line, software or facility components that perform the process functions.
4.3	Allocate processes and product flow Link every process function to its industrial system component with realizedBy. Use allocatedTo, consumes or produces to connect processes to product components where applicable.
4.4	Allocate resources correctly Create Resource elements such as people, energy, tooling or consumables. Link each resource from the industrial system component with requiresResource. Enter required quantity on that relation. Done when: Every process function is realized, and each required resource is attached to the realizing industrial component.

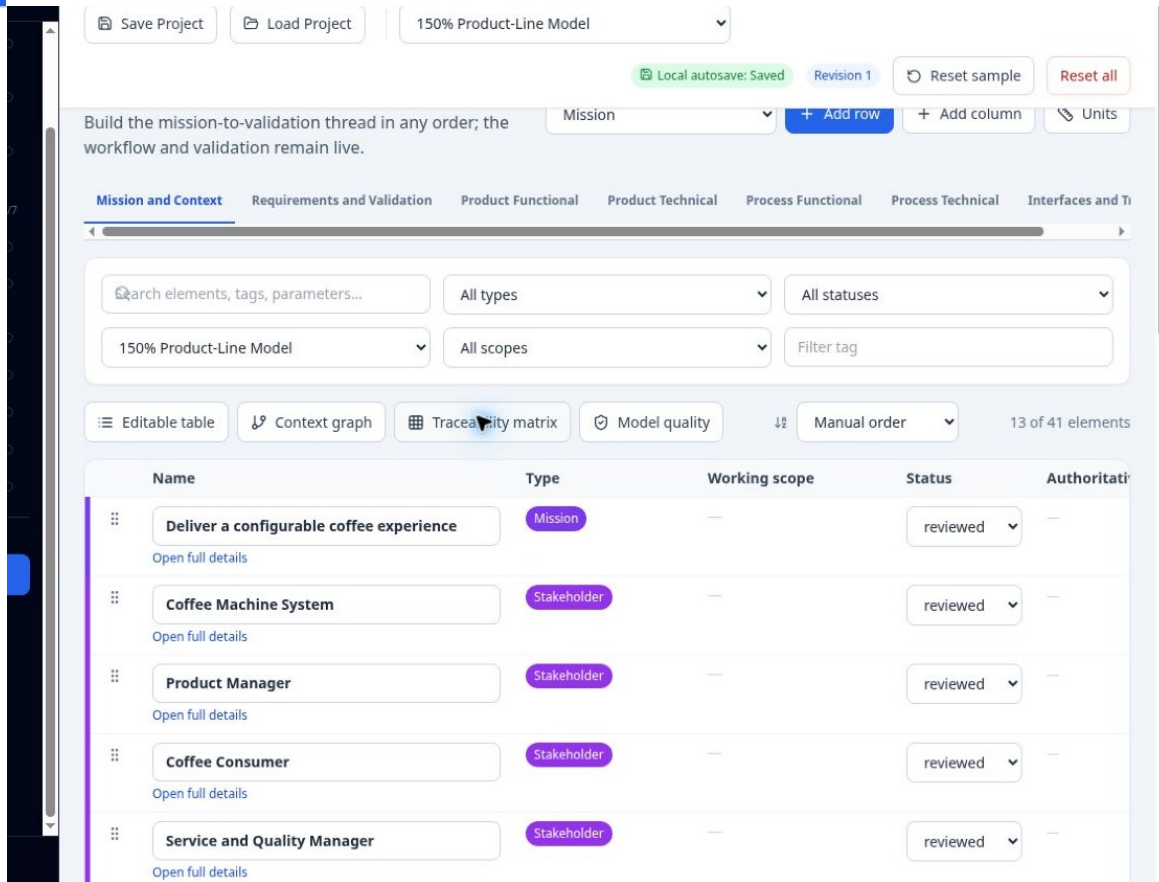
Validate and finish

5.1	Inspect views Use the graph for context, the matrix for cross-domain gaps and Model quality for missing mandatory connections. The table is for editing; the views are for checking.
5.2	Close the trace Check that you can navigate from mission to requirement, to function, to component, and from use case to process function, industrial component and resource.

5.3 Save the evidence

Use Save Project to download a complete project JSON. Optionally export matrices or a PDF report from Export.

Done when: All four dashboard steps are complete and there are no unexplained quality errors.



The editable matrix shows which selected element is connected to which target domain. Use it to link gaps without leaving the view.

Architecture building final check	Expected evidence
Intent	Mission, system of interest, needs and objectives are connected.
Scope	Selected use cases involve the system of interest.
Requirements	Requirements derive from scoped intent and trace to realizing behavior or structure.
Product	Every scoped use case has product functions; every function is realized by product architecture.
Industrial system	Every scoped use case has process functions; every function is realized by industrial system architecture.
Resources	requiresResource connects resources to industrial system components, with quantity where needed.

Chapter 2 - Architecture building + simulation

Use this scope when one architecture must be evaluated with calculated KPIs. Complete every activity in this chapter even if you already read Chapter 1.

Problem space

1	Create mission, system and intent Create the mission, stakeholders, exactly one system of interest, needs and objectives. Link them as described in Chapter 1.
2	Define the use-case and requirement scope Create use cases, link involved stakeholders, select the working scope and derive testable requirements from scoped needs or objectives.
3	Define evaluation KPIs In Parameters and KPIs, create only the measures needed to judge this architecture. Give every KPI a name, direction, unit, weight and objective. Use a standard algorithm when its semantic inputs match your model; otherwise use a formula.

Why define KPIs early

The KPI tells you which engineering inputs the model must contain. Defining it before architecture parameters prevents a simulation with many numbers but no decision value.

The screenshot displays the MBSE / MBPLE Programme workbench interface. On the left is a sidebar with navigation options: Dashboard, PROBLEM SPACE (3/3), SOLUTION SPACE (4/4), and WORKSPACES (Model, Variability, Parameters and KPIs, Simulation, Architecture Trade Study, Export). The main area shows the 'ACTIVE PROJECT' 'Configurable Coffee Machine Product Line' with a dropdown menu set to 'Configurable Coffee Machine Product L'. Below this are buttons for 'Save Project', 'Load Project', and a dropdown for '150% Product-Line Model'. A status bar indicates 'Local autosave: Saved', 'Revision 1', 'Reset sample', and 'Reset all'. The 'PROJECT WORKFLOW' section shows the title 'Configurable Coffee Machine Product Line' and a description: 'Choose the project outcome, then complete the model activities shown in Problem Space and Solution Space.' Below this is the 'Overall scope' section, which includes three options: 'Architecture building' (Not started), 'Architecture building + simulation' (Selected, In progress), and 'Trade study' (Not started). The 'Problem Space' section shows three steps: '1 Define mission, system and intent', '2 Define use-case and requirement scope', and '3 Define evaluation KPIs', with a '3/3 complete' status.

Architecture building + simulation adds KPI definition, architecture parameters and a simulation run.

Solution space - build the architecture

4	Build product functions and product architecture For every scoped use case, create product functions, create product components and connect each function to a realizing component with realizedBy.
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5 Build industrial functions and industrial architecture

Create process functions and industrial system components, connect them with realizedBy, and attach resources to the components with requiresResource.

Solution space - complete architecture parameters

A parameter is a named engineering value stored on a product or industrial component. It is not just a table cell: its type, unit, origin and source make the calculation explainable.

Parameter field	What to enter	Example
Name	A human-readable label.	Equipment mass
Semantic key	A stable meaning used by standard algorithms.	mass, directCost, power
Value and unit	A numeric value and compatible unit.	820 kg
Origin	entered, assumed, calculated or simulated.	assumed
Provenance	Where the value came from.	Supplier preliminary data, 2026-08
Bounds / uncertainty	Expected range or percentage uncertainty.	750-900 kg; 10%
Configurations	Leave empty for all variants or select applicable configurations.	Premium only

6.1 Enter authoritative inputs

Open a product or industrial component and add its parameters. Use consistent units. For process functions, enter duration and its unit. For resource assignments, enter required quantity on the requiresResource relation.

6.2 Create calculated parameters

Choose a calculated numeric parameter, enter a formula using @references, and select the requested output unit. The formula can use +, -, *, /, parentheses, sqrt and powers. It can read parameters on the same component and transitive refiners.

6.3 Check requirement formulas

Bind the requirement symbols to model parameters and enter one comparison. After parameter changes, confirm whether the requirement passes, fails or is pending.

Formula example

If @length and @width are parameters on the same component, area can be @length * @width. A requirement can be @area <= 2.4. Use the formula pickers so the website stores stable IDs behind the friendly symbols.

Save Project Load Project 150% Product-Line Model

Local autosave: Saving Revision 1 Reset sample Reset all

Parameters and KPIs

Editable engineering inputs, exact applicability, safe formulas, units, and standard pre-sizing algorithms.

Engineering inputs KPI definitions

Engineering input table
Blank applicability means every configuration. Value-origin classification and source provenance are independently recorded.

Search owner, name, : + Parameter

Owner / parameter	Semantic key	Value	Range	Origin	Applicability	
Throughput Ingredient-handling module · par-productcomponent-material-handling-module-throughput	throughput	12 parts/h	— ... —	assumed Programme preparation assumption	Every configuration	Edit
Mass Brewing and heating module · par-productcomponent-joining-module-mass	mass	420 kg	300 ... 700	assumed Programme preparation assumption	configuration-manual, configuration-balanced	Edit
Automated mass Brewing and heating module · par-automated-mass	mass	650 kg	500 ... 800	entered Robot-cell supplier budgetary data	configuration-automated	Edit
Manual direct cost Brewing and heating module · par-manual-cost	cost	35000 EUR	25000 ... 50000	assumed Programme preparation assumption	configuration-manual, configuration-balanced	Edit
Automated direct cost Brewing and heating module · par-automated-cost	cost	90000 EUR	70000 ... 120000	entered Robot-cell supplier budgetary data	configuration-automated	Edit

Engineering inputs show process durations and resource quantities used by time, cost and utilization algorithms.

Define KPI formulas

A KPI collects numeric information from anywhere in the active model. The formula uses stable IDs so a renamed element does not break the calculation.

```
param("parameter-id")
```

```
kpi("kpi-id")
```

- Use +, -, *, / and parentheses for arithmetic.
- Use min(...), max(...), sum(...) or average(...) for groups of values.
- Addition and aggregation require exactly matching units. Convert the source values first when units differ.
- A KPI may reference another KPI. Select every dependency in the simulation run.
- The custom columns in tables are descriptive and cannot be used as formula inputs.

7.1 Open the KPI formula builder

Search model parameters by name, unit, element type or tag. Insert the chosen parameter. Do not type an approximate name.

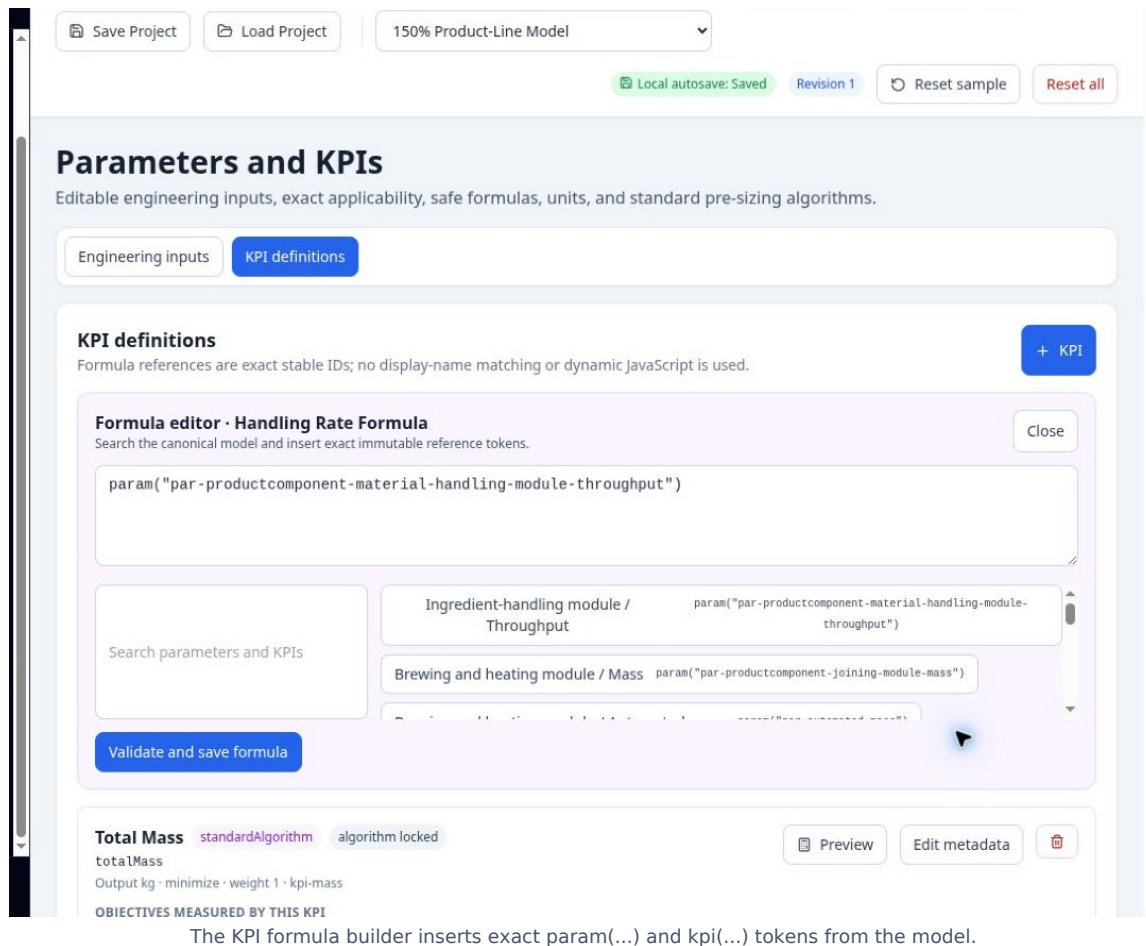
7.2 Set unit, direction and weight

Use Minimize for values such as cost, mass or lead time; Maximize for values such as throughput. Weight expresses importance during a trade study and is normalized with the other selected weights.

7.3 Validate the formula

Resolve missing references and unit errors before simulation. Save the KPI definition.

Done when: Every selected KPI has a reproducible formula or standard algorithm and an output unit.



Run the simulation

8.1	Choose the architecture Open Simulation. Select Architecture-only 150% model for this scope, give the run a meaningful name and choose the KPIs.
8.2	Validate before running Read missing inputs, assumptions and warnings. Correct missing mandatory values. Acknowledge warnings only when the assumption is acceptable for a preliminary engineering estimate.
8.3	Run and inspect Choose Validate and run. Inspect each KPI result, its unit, input sources, warnings and breakdown. Check the affected requirements after the run.
8.4	Record a current result If any model input changes, the old run becomes stale. Run again and use the current result for reporting. Done when: There is a current simulation run containing all selected KPIs with numeric results.

Standard algorithms

The website can calculate total mass, direct element cost, process cost, estimated total cost, total power, manufacturing lead time, resource demand, basic utilization and a throughput proxy. These work only when the expected semantic keys, process durations and resource quantities are present.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

Simulation Saved-run analysis

Run a configured 100% realization or explicitly evaluate the canonical unconfigured 150% model in an architecture context.

Preliminary engineering estimate — not a verified detailed-design result.

Prepare run

ANALYSIS MODE

☒ Configured 100% model
 ☐ Architecture-only 150% model

CONFIGURATION

Essential Capsule · valid

RUN NAME (EDITABLE BEFORE RUNNING)

Essential Capsule — 2026-08-17 09:03

Analysis basis: Essential Capsule · explicit realization available · 4 applied variation effect(s). A background realization is used whenever the explicit one is missing or stale.

KPI definitions

Formula dependencies must also be selected explicitly.

☒ Total Mass kg · kpi-mass

☒ Estimated total cost EUR · kpi-cost

☒ Manufacturing Lead Time h · kpi-lead-time

☒ Resource Demand resource-h · kpi-resource

☒ Throughput Proxy 1/h · kpi-throughput

☒ Total Power kW · kpi-power

Additional direct algorithms

Use these only when no saved KPI definition is needed.

☐ totalMass

☐ directElementCost

☐ processCost

☐ estimatedTotalCost

☐ totalPower

☐ manufacturingLeadTime

Prepare the run by choosing the model, KPIs and a clear run name. Validate warnings before executing.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

Input snapshot.

Premium Dual Boiler — comparable simulation

8/17/2026, 10:23:09 AM · rev 1 Current

Balanced Bean-to-Cup — comparable simulation

8/17/2026, 10:23:09 AM · rev 1 Current

Essential Capsule — comparable simulation

8/17/2026, 10:23:09 AM · rev 1 Current

3 variation point(s) captured · explicit realization.

Preliminary engineering estimate — not a verified detailed-design result. Source project revision 1.

Total Mass 650 kg

KPI kpi-mass

Formula / algorithm totalMass

Input sources Brewing and heating module / Automated mass: 650 kg (Robot-cell supplier budgetary data)

Assumptions None

Missing information None

Warnings None

Estimated total cost 90016.45833333333 EUR

KPI kpi-cost

Formula / algorithm estimatedTotalCost

Input sources Brewing and heating module / Automated direct cost: 90000 EUR (Robot-cell supplier budgetary data); End-of-line test technician: 2 EUR/h (realized process duration 0.0333333333333333 h (Test completed coffee machine) × rate 60 × quantity 1); Coffee-machine integration cell: Automated module assembly cell: 12.08333333333332 EUR/h (realized process duration 0.0833333333333333 h (Integrate coffee-machine modules) × rate 145 × quantity 1); Module assembly workstation: Automated component feeder: 2.375 EUR/h (realized process duration 0.025 h (Manufacture coffee-machine modules) × rate 95 × quantity 1)

Assumptions One engineering day is assumed to equal eight hours.

Missing information None

Warnings None

Manufacturing Lead Time 0.14166666666666666 h

KPI kpi-lead-time

Simulation results retain the numeric result, unit, sources, assumptions, warnings and calculation breakdown.

Architecture + simulation final check	Expected evidence
Architecture	Product and industrial functions are completely realized and traceable.

Architecture + simulation final check	Expected evidence
Inputs	Numeric values have units, origin, provenance and honest uncertainty.
KPI definition	Formula or algorithm, output unit, direction and objective are defined.
Requirements	Numeric requirement formulas are bound and evaluated.
Simulation	The current run contains all selected KPIs and reviewed warnings.
Reproducibility	The result shows which model parameters and assumptions produced it.

Chapter 3 - Trade study

Use this scope when one 150% architecture contains optional or alternative content and the result must be a justified choice between at least two derived 100% architectures.

Problem space

1	Define mission, system and intent Create mission, stakeholders, exactly one system of interest, needs and objectives. Keep the decision context explicit.
2	Define use-case and requirement scope Select the use cases and derive testable requirements from the needs and objectives that matter to the decision.
3.1	Create the trade study question In Architecture Trade Study > Setup, enter a decision name and one question that alternatives can answer. Select the active use cases.
3.2	Select the evaluation intent Select the relevant needs and objectives. Mandatory requirements are collected through their derives traces. Select the KPIs that will be used for every alternative.
3.3	Define variability axes Create one axis per independent choice, such as drive type or capacity class. Give each axis its choices. The guided setup synchronizes each axis to a FeatureGroup under one shared root feature.
One comparison basis All alternatives must be evaluated against the same scoped requirements and the same selected KPIs. Otherwise their scores are not comparable.	

MBSE / MBPLE
Programme workbench

ACTIVE PROJECT
Configurable Coffee Machine Product Line

150% Product-Line Model

Local autosave: Saved Revision 1 Reset sample Reset all

PROJECT WORKFLOW

Configurable Coffee Machine Product Line

Choose the project outcome, then complete the model activities shown in Problem Space and Solution Space.

Overall scope
Changing scope preserves project data and recalculates completion from existing evidence.

☐ Architecture building
Define the problem and build the functional, product and industrial architecture.

☐ Architecture building + simulation
Build one architecture, complete its parameters and simulate it.

☒ Trade study
Build configurable alternatives, simulate architectures, compare and decide.

Status: ☐ Not started ☒ In progress ☐ Complete ☐ Blocked

Problem Space
Define what the system must achieve and the evidence that will judge it.

1 Define mission, system and intent

2 Define use-case and requirement scope

3 Define trade study, variability axes and evaluation KPIs

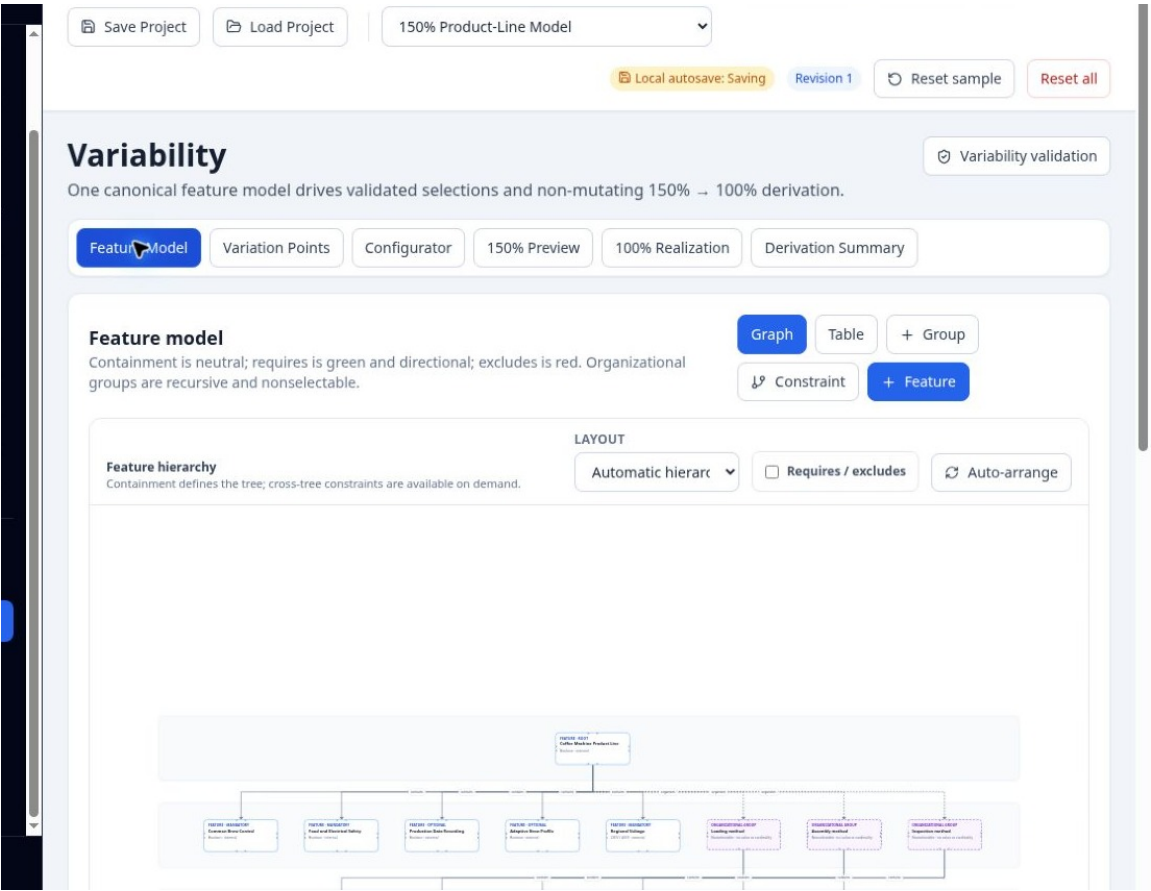
Trade study includes a 150% architecture, feature model, mappings, configurations, 100% derivations, simulations, comparison and decision.

Solution space - Step 1: build the 150% architecture

4.1	Build product functions and product architecture Create common and optional product functions and components. Connect every function with realizedBy. The 150% model contains all content that any considered variant may need.
4.2	Build industrial functions and industrial architecture Create all common and optional process functions, equipment and workstations. Connect processes to industrial components and resources to the realizing components.
4.3	Enter variant-aware parameters Enter common parameters once. For values that apply only to certain variants, select the applicable configurations or map a feature expression to a property modification.

Solution space - Step 2: define feature model and constraints

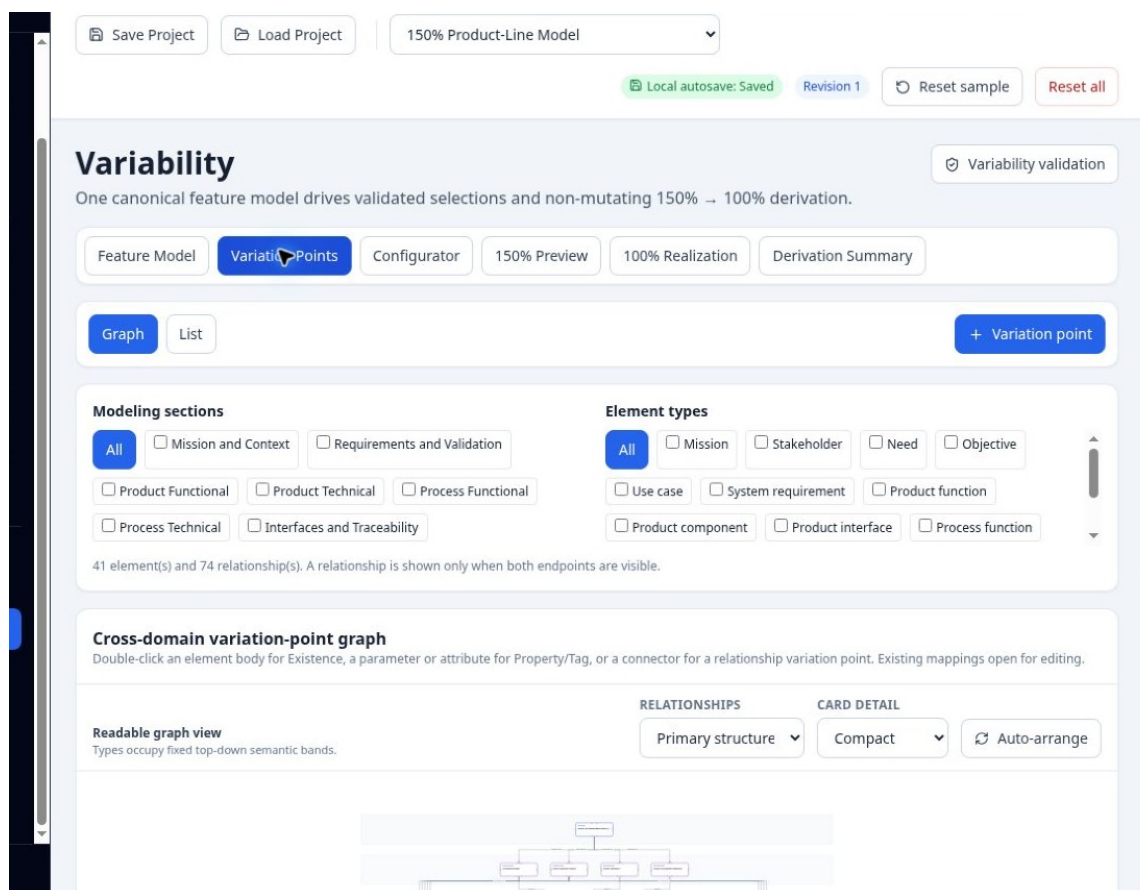
5.1	Create one root In Variability > Feature model, use exactly one root feature for the trade study.
5.2	Create feature groups and features Use mandatory, optional or XOR choices to express valid product and industrial families. Use Boolean or enumeration values when appropriate.
5.3	Add cross-tree rules Add requires when one selection needs another and excludes when two selections cannot coexist. Validate until the feature model has no structural errors. Done when: There is one root and every decision axis is represented by a valid feature or group.



The feature model expresses the legal family of alternatives and cross-tree constraints.

Solution space - Step 3: map variability to architecture

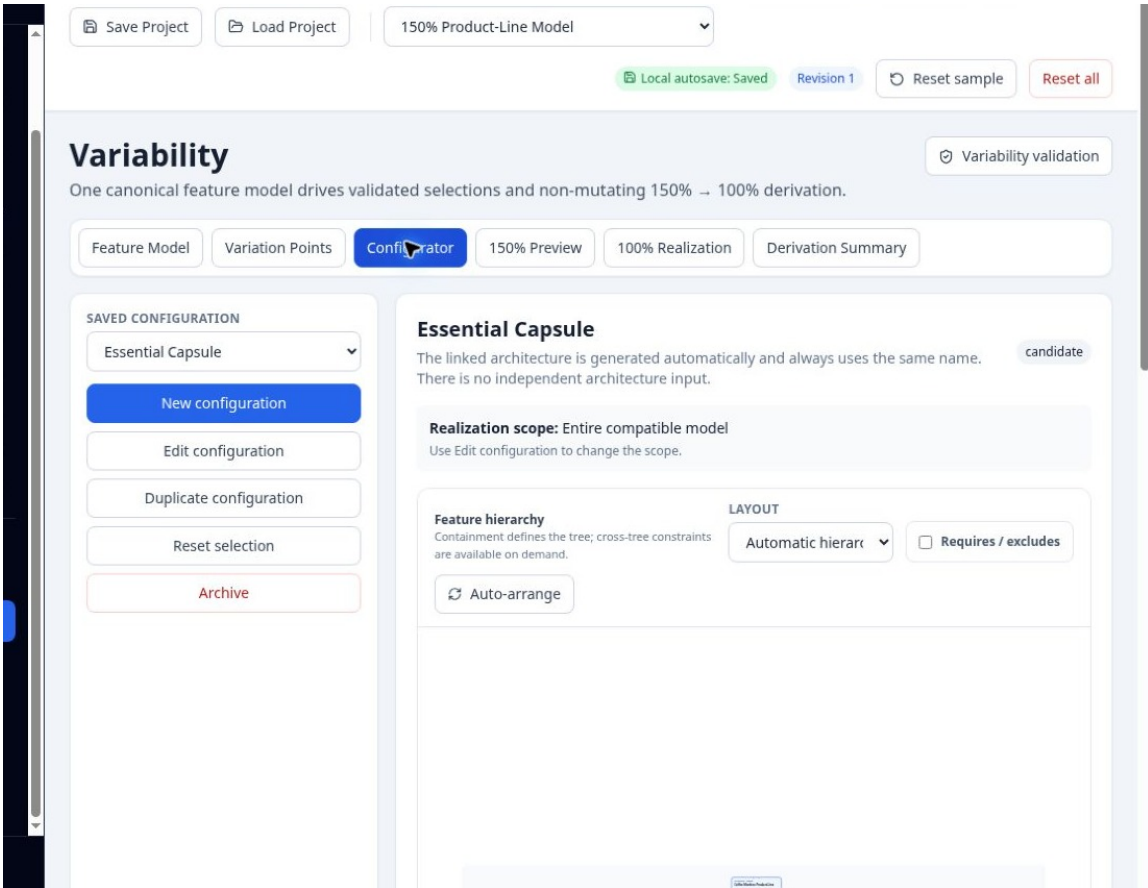
6.1	Create a variation point Choose the feature expression that controls the model content.
6.2	Choose the target Map the expression to an architecture element, relationship or property. Use element mappings to include or suppress parts, relationship mappings to include or suppress links, and property mappings to change a value.
6.3	Preview the 150% status Inspect common, included, excluded and modified previews. Correct mappings that would orphan a function or component. Done when: Every feature that changes the architecture has an intentional mapping.



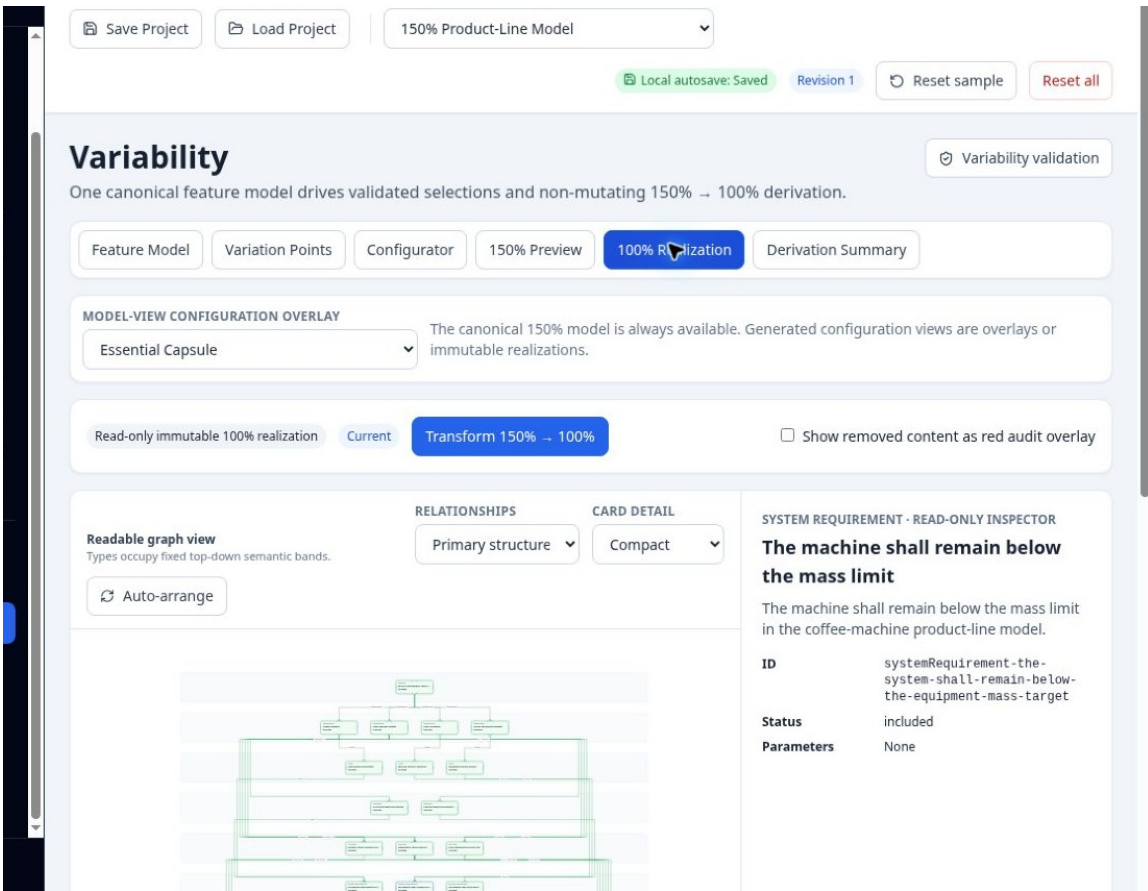
Variation points connect feature expressions to architecture elements, relations or properties.

Solution space - Steps 4 and 5: configure and derive

7.1	Create at least two configurations Open Configurator, create a named configuration and select features. Let requires rules add dependent selections. Resolve excludes and group errors until the configuration is valid.
7.2	Repeat for a meaningful alternative The second configuration must represent a real decision alternative, not a duplicate label.
8.1	Derive a 100% architecture Select a valid configuration and choose Derive / realize. Read warnings and acknowledge only understood preliminary assumptions.
8.2	Inspect the realization Review included, excluded and modified elements and relations. The realization is read-only and records the 150% source revision and audit.
8.3	Keep it current A later source-model or mapping change makes the realization stale. Derive again before simulation or comparison. Done when: At least two non-archived configurations have explicit current 100% realizations.



A configuration is usable only when feature selections and constraints are valid.



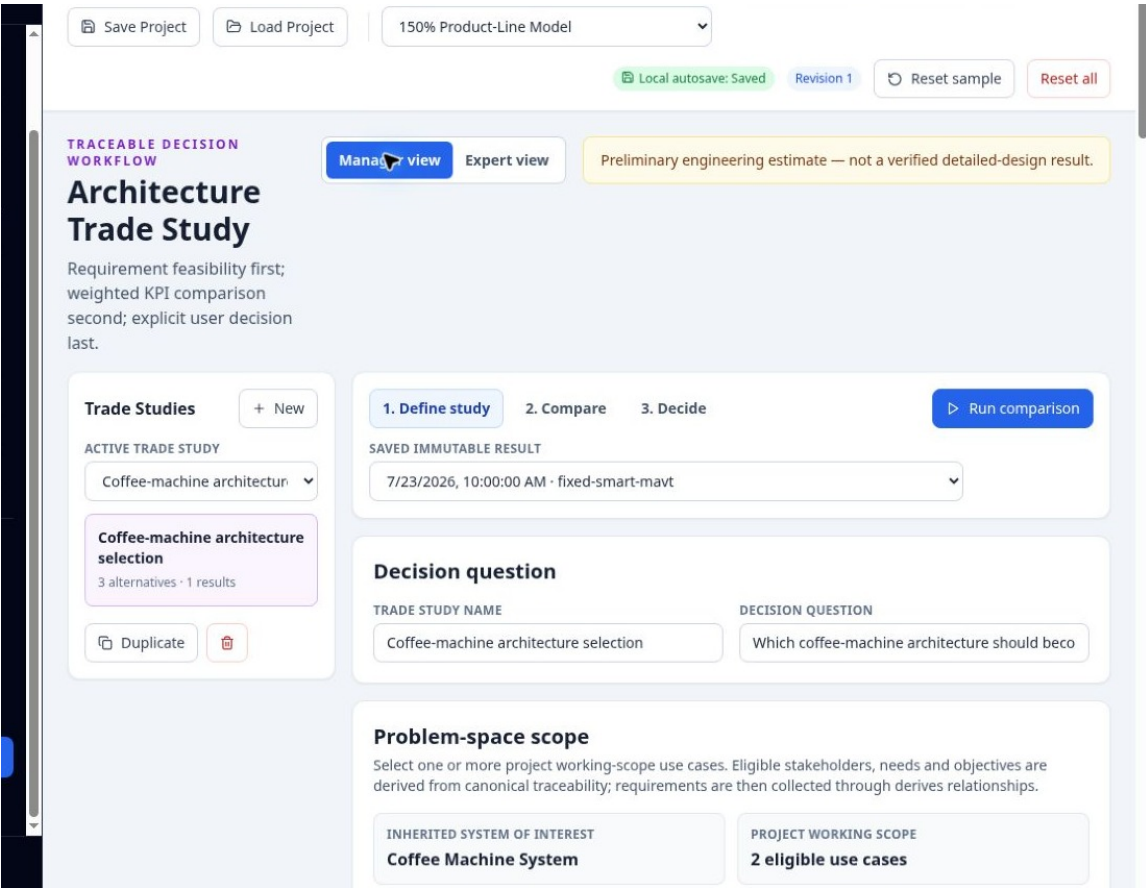
A 100% realization records what was included, excluded or modified and which 150% revision produced it.

Solution space - Step 6: simulate every alternative

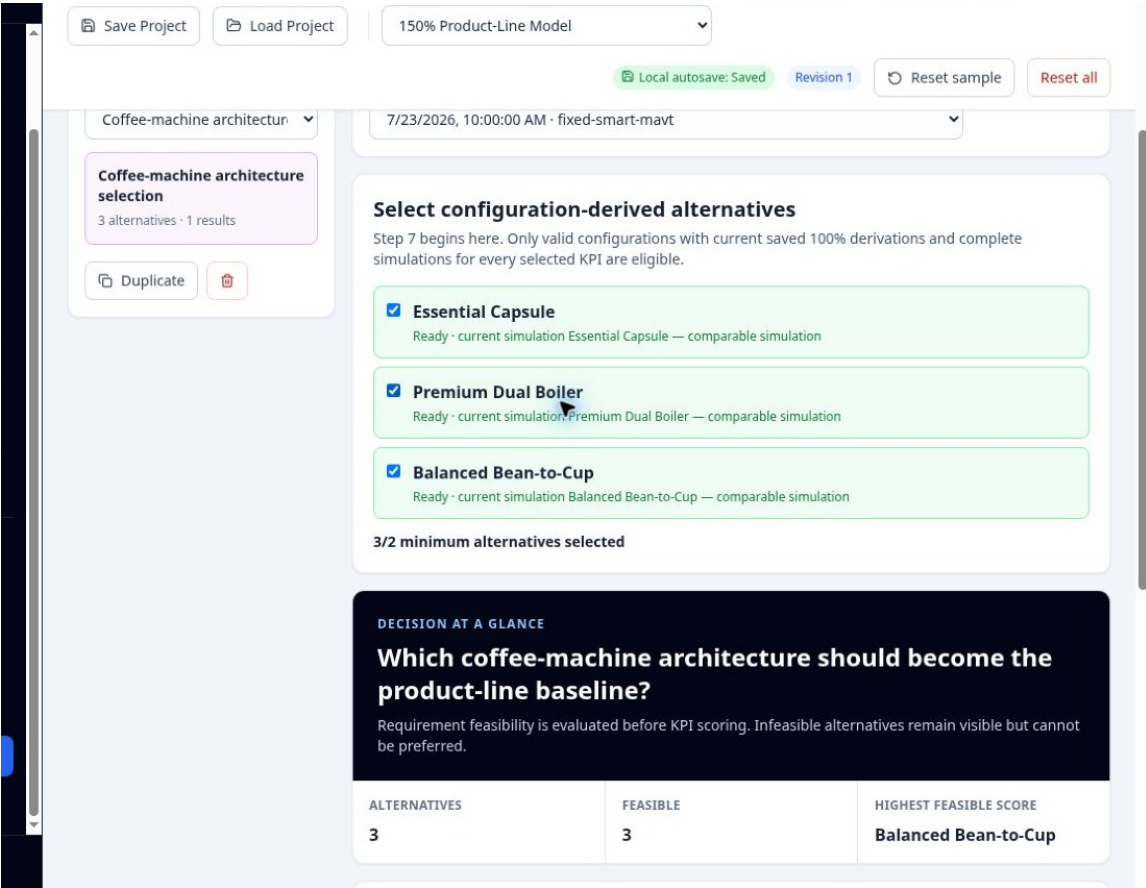
9.1	Complete variant inputs For each configuration, check parameter applicability, process durations and resource quantities. Use identical units and KPI definitions.
9.2	Run the same KPI set In Simulation, choose Configured 100% model, select the first current realization and all trade-study KPIs. Validate and run. Repeat for every alternative.
9.3	Review requirement evidence A run can be numerically complete but infeasible. Inspect requirement actual values versus thresholds and resolve pending bindings.
9.4	Check current status If a realization or input changes, rerun. Only a current complete simulation makes the alternative eligible.

Solution space - Step 7: compare and decide

10.1	Select eligible alternatives In Trade Study > Alternatives, select only configurations with a current explicit realization and a current complete simulation containing every selected KPI.
10.2	Review feasibility first In Manager view, inspect requirement feasibility. Failed alternatives remain visible but cannot win the comparison.
10.3	Review KPI ranking second Inspect raw values and normalized weighted scores. In Expert view, inspect weights, contributions, formulas, provenance and requirement evidence.
10.4	Make the decision explicitly Create a decision from the current comparison, choose a feasible alternative, enter owner and rationale, confirm baseline approval and approve. Do not treat the highest score as an automatic decision. Done when: An approved decision references a current comparison and a feasible selected alternative.



Setup fixes the decision question, scope, variability axes, objectives, requirements and shared KPI set.



Manager view presents feasibility and ranking for eligible alternatives.

Save ProjectLoad Project

150% Product-Line Model

Local autosave: SavingRevision 1Reset sampleReset all

Alternative	KPI	Raw evidence	Normalized	Global weight	Contribution	Formula and provenance
Essential Capsule	Total Mass	420.00 kg	100.00	1	160.00	420 transform 80; raw weight 4e73-a281-60c90cfa8f4-66a4 totalMass.
Essential Capsule	Estimated total cost	35010.25 EUR	100.00	1	268.39	35010.25 transform function to 89. a3f8da54-edcc derivation-90c KPI formula est
Essential Capsule	Manufacturing Lead Time	0.18 h	0.00	1	200.00	0.1833333333 minimize value simulation-a3f 6053efd5106b 453c-85fa-44a manufacturing
Essential Capsule	Throughput Proxy	5.45 1/h	0.00	1	43.64	5.4545454545 value function simulation-a3f 6053efd5106b 453c-85fa-44a throughputPro
Balanced Bean-to-Cup	Total Mass	420.00 kg	100.00	1	160.00	420 transform 80; raw weight 47d5-bb67-22. 8053a047-06e formula totalM
Balanced Bean-to-Cup	Estimated total cost	35016.46 EUR	99.99	1	268.37	35016.458333 value function simulation-45f 22ab379ccb91 06eb-4bbb-9b estimatedTotal
Balanced	Manufacturing	0.14 h	100.00	1	200.00	0.1416666666

Expert view exposes raw and normalized values, weights, contributions, formulas and provenance.

Save ProjectLoad Project

150% Product-Line Model

Local autosave: SavedRevision 1Reset sampleReset all

TRACEABLE DECISION WORKFLOW

Manager viewExpert view

Preliminary engineering estimate — not a verified detailed-design result.

Architecture Trade Study

Requirement feasibility first; weighted KPI comparison second; explicit user decision last.

Trade Studies

+ New

ACTIVE TRADE STUDY

Coffee-machine architecture

Coffee-machine architecture selection

3 alternatives · 1 results

Duplicate

1. Define study2. Inspect evidence3. Decide

Run comparison

SAVED IMMUTABLE RESULT

7/23/2026, 10:00:00 AM · fixed-smart-mavt

Decision and rationale

approved

Only requirement-feasible alternatives are eligible. One feasible alternative is sufficient, with an explicit no-competitive-ranking notice.

SELECTED FEASIBLE ALTERNATIVE

Balanced Bean-to-Cup

OWNER

Sample Product Governance Board

RATIONALE

Sample governance decision: mandatory feasibility, Pareto position, fixed stakeholder value, sensitivity and bounded robustness were reviewed together.

The final step records the selected feasible alternative, owner, rationale and approval.

Trade study final check	Expected evidence
Decision scope	Question, use cases, intent, requirements and KPIs are fixed.

Trade study final check	Expected evidence
150% architecture	All common and variable product and industrial content is connected.
Variability	One root, valid groups and constraints, and architecture mappings.
Alternatives	At least two valid configurations with current explicit 100% realizations.
Evaluation	Every alternative has a current complete run with the same KPI set.
Feasibility	Requirement evidence is evaluated before ranking.
Decision	A feasible alternative is explicitly approved with owner and rationale.

Quick reference - formulas and standard calculations

Requirement formulas

Use arithmetic and exactly one comparison: <, <=, >, >=, =, == or !=. Insert bound symbols with the picker. Missing values leave the requirement pending or failed; they do not silently pass.

```
@equipment_mass <= 1000
```

Calculated parameter formulas

Use @references, +, -, *, /, parentheses, sqrt and powers (^ or **). References must be on the same component or a transitive refiner. Select a requested output unit when conversion is needed.

```
@length * @width * @height
```

KPI formulas

Use exact stable tokens from the formula builder. A formula can reference model parameters and other KPIs.

```
sum(param("mass-a"), param("mass-b"))
```

Standard KPI	Information the model must contain	Meaning
Total mass	Active numeric parameters with semantic key mass.	Sum of mass inputs in the active model.
Direct element cost	Active numeric parameters with semantic key directCost.	Sum of direct component costs.
Process cost	Process duration, realizing industrial component, required resource quantity and resource hourlyRate.	Duration x rate x quantity, summed.
Estimated total cost	Direct element cost and process cost inputs.	Direct + process cost.
Total power	Active numeric parameters with semantic key power.	Sum of power inputs.
Manufacturing lead time	Process duration and precedes relations.	Longest path through the process network; parallel branches use the critical path.
Resource demand	Process duration and requiresResource quantity.	Resource-hours with per-resource breakdown.
Basic utilization	Resource demand, capacity hours and availability percent.	Demand divided by effective capacity.
Throughput proxy	Manufacturing lead time.	1 / lead-time hours; useful for comparison, not detailed production planning.

Time convention

For the standard algorithms, 1 day equals 8 hours. Confirm that this preliminary convention is acceptable for your study and record any differing assumption.

Saving, reviewing and correcting

Save safely

- Local autosave protects work in the same browser profile, but it is not a collaboration server or durable backup.
- Save Project downloads the complete portable JSON. Use a clear filename after each major milestone.
- Load Project validates the file. Load as new to preserve the current project; replace only when intended. The website makes a safety snapshot before replacement.
- A model change can make derivations, simulations or comparisons stale. Recreate only the affected downstream evidence.

When a dashboard step is blocked

Symptom	Likely cause	Correction
Mission step incomplete	No hasStakeholder trace, no unique system of interest, or missing need/objective.	Fix Mission and Context and flag exactly one system of interest.
Scope incomplete	Selected use case does not involve system of interest or no requirement derives from scoped intent.	Add involvedIn and derives traces.
Architecture incomplete	Use case lacks a function or a function lacks a realizing component.	Add hasFunction and realizedBy traces.
Simulation blocked	Missing input, unit error, unresolved KPI dependency or unacknowledged warning.	Open validation details and correct the named source.
Alternative ineligible	Invalid configuration, stale/missing realization, stale run or missing selected KPI.	Validate, rederive and rerun in that order.
Trade result stale	Study settings, model, derivation or simulation changed.	Refresh eligible alternatives and create a new current comparison.

Final habit

Do not ask only whether a number looks right. Ask which element supplied it, which unit and assumption were used, which requirement it affects and whether the result is current.

Architect and Modeler answer handbook - first-use edition

This branch-accurate handbook supplements the established guide. It explains what an unfamiliar user should enter, what the answer creates or changes, and when a question is review-only. Dynamic questions repeat for each item in the project.

Handbook index

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How to answer Architect questions

The number of visible questions changes with the chosen scope and with the items you create. Questions marked "per item" repeat for each stakeholder, use case, requirement, function, component, KPI, feature axis or alternative.

Study setup and intent (AV-A01 to AV-B07)

Question	What to enter or select	What it changes
AV-A01 - project aim	Describe the intended project outcome and why the work is being undertaken.	Stores the authoritative project description before scope selection.
AV-A02 - study scope	Choose Architecture definition, Architecture and simulation, or Trade-off according to the decision you need to support.	Sets how far the guided workflow continues. Existing project data is preserved when scope changes.
AV-A03 - study name	A short unique project name, such as Office coffee-machine concept.	Renames the active project.
AV-A04 - mission	One clear sentence describing the system's purpose and the problem it solves. Avoid listing a preferred solution.	Creates or updates the Mission element.
AV-B01 - stakeholders	Semicolon-separated people or organisations whose concerns matter. Do not enter the system being designed as a stakeholder.	Creates Stakeholder elements and canonical mission-participation links.
AV-A05 - system of interest	Enter the name of the system being designed or analysed.	Creates or updates the dedicated System element and its canonical hasSOI link from the Mission; use cases refer to the System as their subject.
AV-B06/B07 - external systems and roles	Enter external-system names, then describe each system's role or boundary interaction.	Creates External System elements and canonical Mission participatesInMission links; those systems can later receive involvedIn links to use cases.
AV-B02 - stakeholder role	Describe why each stakeholder matters.	Updates documentation only; it creates no relationship and changes no calculation.
AV-B03 - stakeholder needs	Desired outcomes or problems from that stakeholder's viewpoint, separated by semicolons.	Creates Need elements linked to the stakeholder.
AV-B04 - stakeholder objectives	Measurable improvements or targets, separated by semicolons.	Creates Objective elements linked to the stakeholder.
AV-B05 - intent recap	Confirm only after mission, stakeholders, needs and objectives describe the study correctly.	Records review; it does not override validation findings.

Use cases and working scope (AV-C01 to AV-C05)

Question	What to enter or select	What it changes
AV-C01 - use cases	Short action-oriented scenarios, separated by semicolons, such as Prepare beverage; Perform maintenance.	Creates Use Case elements.
AV-C02 - participants	For each use case, select all involved stakeholders, including the system of interest where applicable.	Creates use-case participation links.
AV-C03 - addressed intent	Select only the needs/objectives actually addressed by that use case.	Creates canonical addresses relationships.
AV-C04 - analysis scope	Select the use cases this study will analyse. Only eligible cases involving the system of interest are offered.	Sets the persistent working use-case scope.
AV-C05 - scope recap	Check participants, addressed intent and active cases.	Records review and exposes missing scope evidence.

Requirements (AV-D01 to AV-D10)

Question	What to enter or select	What it changes
AV-D01/D02 - requirements from needs/objectives	Testable statements. Reuse an existing requirement when it expresses the same obligation; otherwise add a new one.	Creates System Requirement elements and canonical derives links.
AV-D03 - relevant use cases	Select the active use cases in which each requirement applies.	Stores guided applicability context.
AV-D04 - quantitative or qualitative	Select quantitative when a number and comparison can test the requirement; qualitative when inspection, demonstration or another non-numeric judgement is appropriate.	Chooses the formula or verification path.
AV-D05 - quantitative check	Enter measured property, semantic key, operator, target and unit. Example: mass, semantic key mass, <=, 12, kg.	Creates a formal requirement formula intent. It does not invent an actual value.
AV-D06 - actual-value source	Select the function/component that owns the measured property, or leave ownership for later architecture work.	Binds the formula intent to an authoritative model owner when one exists.
AV-D07 - preliminary value	Enter a value only when supported; include unit, source, origin and optional uncertainty/range. Otherwise leave it pending.	Creates or updates the owner's parameter and evidence fields.
AV-D08 - qualitative verification	Name an analysis, inspection, demonstration or test and describe what will be checked.	Creates or updates a Verification Method.
AV-D09 - importance	Mandatory means failure makes an alternative infeasible; Important and Desirable express lower decision priority.	Sets the requirement class.
AV-D10 - requirement recap	Confirm origins, wording, formula/verification path and ownership.	Records review without forcing a passing result.

Trade framing and early KPI intent (Trade-off only: AV-T01 to AV-T06, then AV-I01 to AV-I07)

Question group	What to enter or select	What it changes
AV-T01/T02 - study and decision	A short Trade Study name and one explicit decision question comparing feasible concepts.	Creates/updates the active Trade Study and its open decision context.
AV-T03 - common content	Capabilities or solution elements required in every alternative; exclude things that vary.	Proposes mandatory common features.
AV-T04 - variation areas	Independent choice areas such as brewing technology or interface type, not whole alternatives.	Proposes reusable variability axes.
AV-T05 - concepts	At least two alternative names with short descriptions.	Creates configuration intents used later.
AV-T06 - framing recap	Verify decision, common content, variation axes and concepts.	Records review.
AV-I01 - evaluation results	Select only KPIs needed to judge the objectives.	Creates/selects canonical KPI definitions; it does not run analysis.
AV-I02 - KPI purpose	Link each KPI to the needs/objectives it measures.	Updates KPI objective links.
AV-I03 - calculation method	Use a standard algorithm when its meaning and inputs fit; otherwise choose a guided formula.	Sets the KPI calculation mode.
AV-I06 - interpretation	Enter output unit, minimize/maximize direction, target and optional acceptable bounds.	Defines how the KPI is read.
AV-I07 - global weight	Enter a finite non-negative weight. At least one selected KPI must be greater than zero.	Sets the reusable KPI default weight.

Product behavior and product architecture (AV-E01 to AV-F06)

Question group	What to enter or select	What it changes
AV-E01/E02 - product functions and order	For each use case, enter product functions in normal execution order; then correct the proposed sequence if needed.	Creates Product Function elements, use-case hasFunction links, a contextual sequence and cycle-free precedes links.
AV-E03 - function satisfaction	Select requirements directly helped by each function; leaving a function unlinked is allowed when justified.	Creates requirement satisfiedBy links.
AV-E04 - product-behavior recap	Check use-case coverage, sequence and requirement links.	Records review.
AV-F01 - realizing components	Name one or more physical/logical product components that perform each function.	Creates Product Component elements and realizedBy links.
AV-F02 - component satisfaction	Select requirements directly satisfied by each component.	Creates requirement satisfiedBy links.
AV-F03/F04 - parameter owner/value	Select the authoritative product function/component for a pending measured property; then enter value, unit, source and origin if known.	Creates or reuses one canonical parameter and binds the requirement formula.
AV-F05 - product interaction	When useful, select another product component, name the interface and describe exchanged material, energy or information.	Creates Interface content and canonical links.
AV-F06 - product-architecture recap	Check function realization, interfaces, parameters and satisfaction.	Records review and exposes gaps.

Industrial behavior and industrial architecture (AV-G01 to AV-G12)

Question group	What to enter or select	What it changes
AV-G01/G02 - process functions and order	For each use case, enter industrial-system process functions such as positioning, fastening or inspection, then verify their order. Leave blank for a genuinely product-only case.	Creates Process Function elements, hasFunction, sequences and precedes links.
AV-G03 - process satisfaction	Select requirements each process function helps satisfy.	Creates requirement satisfiedBy links.
AV-G04 - realizing equipment	Name the workstation, equipment or plant element that performs each process function.	Creates Industrial System Components and realizedBy links.
AV-G05/G05A/G05B - industrial evidence	Select satisfied requirements; resolve any property owned by an industrial function/component and enter evidence if known.	Creates satisfaction links and canonical parameters.
AV-G06 - duration	Enter the authoritative process duration, time unit, source, origin and optional uncertainty.	Updates process metadata used by lead-time, cost and resource calculations.
AV-G07/G08 - consumed/produced product	Select product components and enter positive quantity and unit.	Creates canonical consumes and produces flows.
AV-G09/G10 - resources	Name required people, roles, skills, tools, machines, software or facilities; then enter quantity, unit, optional rate, capacity and availability.	Creates Resource elements and industrial-component requiresResource links with quantitative metadata.
AV-G11 - industrial interaction	Add only meaningful material, energy or information exchanges between industrial components.	Creates interface content and links.
AV-G12 - industrial recap	Check realization, sequence, handoffs, duration and resources.	Records review and exposes model-quality gaps.

Traceability and readiness (AV-H01 to AV-H04)

Question	What to enter or select	What it changes
AV-H01 - verification	For every requirement, identify a credible analysis, inspection, demonstration or test; optionally allocate a process function.	Creates/updates Verification Methods and verifies/allocation links.
AV-H02 - satisfying evidence	Select at least one compatible function, technical component or existing owned parameter. Do not create a second parameter here.	Creates satisfaction links or binds the selected parameter and its owner to the requirement formula.
AV-H03 - evidence credibility	Confirm only when value, target, unit, source, origin and verification are credible; otherwise mark pending.	Records an explicit evidence-review state.
AV-H04 - architecture readiness	Review intent, behavior, realization, satisfaction, flows, resources and verification.	Calculates readiness from the model; confirmation cannot hide blockers.

Simulation questions (Architecture and simulation: AV-I01 to AV-I12)

After choosing KPIs and their purpose/calculation/interpretation as described above:

Question group	What to enter or select	What it changes
AV-I04/I05 - exact calculation	Confirm the inputs used by a standard algorithm, or build a formula by inserting named parameters/KPIs.	Stores no duplicate values; formulas keep immutable IDs.
AV-I08 - missing inputs	Select every authoritative contributing owner and enter separate value, unit, source, origin and optional uncertainty.	Creates/updates canonical parameters used by the KPI.
AV-I11 - run	Enter a meaningful run name and execute.	Validates inputs and creates one immutable Simulation Run.
AV-I12 - Simulation Results	Confirm only after KPI results, requirement status, warnings and sources match the intended analysis.	Records review of current evidence. The current architecture is selected automatically.

Variability and 100% configurations (Trade-off only: AV-J01 to AV-K05)

Question group	What to enter or select	What it changes
AV-J01/J02 - family and common features	Name the complete product line and confirm mandatory capabilities shared by all alternatives.	Creates one Root Feature and mandatory child features.
AV-J03/J04 - axes and choices	For each variation area, list choices and select XOR, OR, optional or typed mode; list allowed values for typed axes.	Creates synchronized Variability Axes, FeatureGroups and Features.
AV-J05 - constraints	Add only necessary requires or excludes rules.	Prevents invalid feature selections.
AV-J06 - element applicability	Map only genuinely variable requirements/functions/components/processes/resources/verification methods to a feature; unlisted elements remain common.	Creates existence applicability using the existing variation mechanism.
AV-J07/J08 - property variation and domain	Map a feature to a changed existing parameter value and select affected model domains when needed.	Creates typed Variation Points with exact target and realization scope.
AV-J09 - 150% recap	Check family completeness, hierarchy, constraints, applicability and property changes.	Records review.
AV-K01 - configuration	For each concept, choose one value for every required group and any optional features.	Creates/updates a Configuration.
AV-K02/K03 - invalid selection	Apply a proposed required-feature correction or return to the choices.	Keeps invalid drafts visible but blocks derivation/simulation until valid.
AV-K04 - derive	Review included, excluded and modified content, then derive.	Creates immutable 100% realization evidence without changing the 150% source.
AV-K05 - realization recap	Compare selected features with included/excluded elements and applied property changes.	Confirms the current derivation.

Simulate, compare and decide (Trade-off only: AV-S01 to AV-L10)

Question group	What to enter or select	What it changes
AV-S01 - readiness	Verify every intended alternative has a valid configuration and current 100% derivation.	Identifies blockers; it creates no run.
AV-S03/S04 - comparable simulation	Run the same KPI set for every alternative, then review coverage, units, warnings and sources side by side.	Atomically creates one immutable configured Simulation Run per alternative; no partial batch is stored.
AV-L01/L02/L03 - comparison set	Select at least two eligible alternatives, mandatory feasibility requirements and comparable KPIs.	Sets the active study's evidence basis.
AV-L04/L05 - priorities and limits	Set study-specific KPI weights, preferred direction and optional warning/hard thresholds.	Updates study settings without changing global KPI definitions.
AV-L06/L07/L08 - compare	Resolve blocking comparability gaps, run the calculation and review feasibility, weighted scores, ties and sensitivity.	Stores an immutable Comparison Result. Infeasible alternatives remain visible but cannot be preferred.
AV-L09/L10 - decision	Explicitly select a feasible baseline, record rationale, owner, status/date and confirm approval.	Creates/updates the Decision and changes the baseline only after explicit approval.

Final overview (AV-M02 to AV-M06)

The final pages are reviews, not new data-entry steps. They summarise the study question, model content, stakeholder traceability, requirements, architecture, current simulation and - for Trade-off - alternatives and decision. The completion banner is green only when required answers and evidence are complete and reviewed; amber means pending evidence, warnings, skipped/incomplete answers or answers needing review; red means a failed requirement or another blocking model finding.

Using Modeler view safely

Use Project Workflow to find the next activity. Each Model section edits canonical elements and relationships through Table, Graph, Sequence, Matrix, Section Recap, Model Digital Thread and Model Quality views as applicable. Sequence overview displays the actual precedes network, including parallel branches and joins. The table keeps core fields visible and reveals status, tags, custom attributes and connections through **More columns**. Element editing always opens in the right-side drawer. Interfaces and Traceability are separate sections. Variability edits the feature model, variation points, configurations and 150%/100% realizations; organizational feature groups appear in the same hierarchy table as features. Parameters and KPIs edits authoritative engineering inputs and formulas. Simulation creates immutable runs. Architecture Trade Study frames, compares and decides. Project Recap follows the complete model from mission through architecture, evidence and decision. Delivery saves projects, snapshots and exports. Model Quality collects project-wide findings and links back to affected content.

Section recaps do not prescribe arbitrary element counts. They derive their expectations from current model rules and upstream content-for example, every defined function needs a realizing component and every selected use case needs behavior. Guided examples may state expected counts as walkthrough aids, but those counts are not project validation rules.

When an Architect question is unclear, switch to Modeler view to inspect the relevant table, matrix or graph, then return. If a Modeler edit changes content used by a confirmed answer, the affected question and recap are marked for review; unrelated documentation edits stay confirmed. Never duplicate an existing parameter merely to satisfy a question: identify its authoritative owner and reuse the canonical value.

Quick answer-quality checklist

- Use stakeholder language for needs and measurable language for objectives.
- Write requirements so a verification method or formula can decide them.
- Give every numeric value a unit, source and value origin.
- Keep product functions distinct from the product components that realize them.
- Keep industrial process functions distinct from workstations/equipment that realize them.
- Allocate resources to the industrial component that requires them.
- Treat optional content and changed values as variability; keep common content unconditioned.
- Compare only current 100% derivations with current, complete and comparable simulation evidence.
- Record a decision rationale; do not treat the highest score as automatic approval.